

HW 0331

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5.

(a)

$$\begin{aligned}\sum_{t=1}^T (t - \bar{t})^2 &= \frac{T(T+1)(2T+1)}{6} - \frac{[\frac{T(T+1)}{2}]^2}{T} \\ \sum_{t=1}^T (t - \bar{t})^2 &= \frac{T(T^2 - 1)}{12} \\ \text{Var}(\hat{\beta}_2 | \mathbf{x}) &= \frac{\sigma^2}{\sum_{t=1}^T (x_t - \bar{x})^2} = \frac{\sigma^2}{\frac{T(T^2-1)}{12}} = \frac{12\sigma^2}{T^3 - T} \\ \lim_{T \rightarrow \infty} \frac{12\sigma^2}{T^3 - T} &= 0\end{aligned}$$

Thus, the least squares estimator is consistent.

(b)

$$\begin{aligned}\sum x_t &= \sum (0.5)^t \rightarrow \frac{0.5}{1 - 0.5} = 1 \\ \sum x_t^2 &= \sum (0.25)^t \rightarrow \frac{0.25}{1 - 0.25} = \frac{1}{3} \\ \sum (x_t - \bar{x})^2 &= \sum x_t^2 - \frac{(\sum x_t)^2}{T} \approx \sum x_t^2 \rightarrow \frac{1}{3} \text{ as } T \rightarrow \infty \\ \text{Var}(\hat{\beta}_2 | \mathbf{x}) &= \frac{\sigma^2}{\sum (x_t - \bar{x})^2} \rightarrow \frac{\sigma^2}{1/3} = 3\sigma^2 \\ \lim_{T \rightarrow \infty} \text{Var}(\hat{\beta}_2 | \mathbf{x}) &= 3\sigma^2 \neq 0\end{aligned}$$

Thus, the least squares estimator is NOT consistent.

(c)

In (a), the X_t become more dispersed as T increases, so OLS learns the slope precisely.

In (b), the X_t rapidly clumps near 0, the extra data add little information, and the OLS cannot estimate the slope consistently.

6.

(a)

Under 5% significance level, $t_c = t_{0.025, 60} \approx 2.000$

$$t = \frac{b_2 - 0}{se(b_2)} = \frac{3}{\sqrt{4}} = 1.5$$

$$|1.5| < 2.000$$

DO NOT reject null hypotheses $\beta_2 = 0$.

(b)

$$\begin{aligned}\hat{g} &= b_1 + 2b_2 = 2 + 2(3) = 8 \\ \widehat{Var}(\hat{g}) &= \widehat{Var}(b_1) + 4\widehat{Var}(b_2) + 2(1)(2)\widehat{Cov}(b_1, b_2) \\ &= 3 + 4(4) + 4(-2) \\ &= 3 + 16 - 8 \\ &= 11\end{aligned}$$

$$\begin{aligned}se(\hat{g}) &= \sqrt{11} \approx 3.317 \\ t &= \frac{\hat{g} - 5}{se(\hat{g})} = \frac{8 - 5}{3.317} = \frac{3}{3.317} \approx 0.904 \\ |0.904| &< 2.000\end{aligned}$$

DO NOT reject null hypotheses $\beta_1 + 2\beta_2 = 5$.

(c)

$$\begin{aligned}\hat{h} &= b_1 - b_2 + b_3 = 2 - 3 + (-1) = -2 \\ \widehat{Var}(\hat{h}) &= Var(b_1) + Var(b_2) + Var(b_3) - 2Cov(b_1, b_2) + 2Cov(b_1, b_3) - 2Cov(b_2, b_3) \\ &= 3 + 4 + 3 - 2(-2) + 2(1) - 2(0) \\ &= 10 + 4 + 2 = 16\end{aligned}$$

$$\begin{aligned}se(\hat{h}) &= \sqrt{16} = 4 \\ t &= \frac{\hat{h} - 4}{se(\hat{h})} = \frac{-2 - 4}{4} = \frac{-6}{4} = -1.5 \\ |-1.5| &< 2.000\end{aligned}$$

DO NOT reject null hypotheses $\beta_1 - \beta_2 + \beta_3 = 4$.

20.

(a)

$$\begin{aligned}\bar{y}_2 &= \beta_1 + \beta_2\bar{x}_2 + \bar{e}_2, \quad \bar{y}_1 = \beta_1 + \beta_2\bar{x}_1 + \bar{e}_1 \\ \hat{\beta}_{2,\text{mean}} &= \frac{(\beta_1 + \beta_2\bar{x}_2 + \bar{e}_2) - (\beta_1 + \beta_2\bar{x}_1 + \bar{e}_1)}{\bar{x}_2 - \bar{x}_1} = \beta_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1} \\ E[\hat{\beta}_{2,\text{mean}}|\mathbf{x}] &= \beta_2 + \frac{E[\bar{e}_2|\mathbf{x}] - E[\bar{e}_1|\mathbf{x}]}{\bar{x}_2 - \bar{x}_1} = \beta_2 \text{ by MR4}(E[e_i|\mathbf{x}] = 0)\end{aligned}$$

Thus, $\hat{\beta}_{2,\text{mean}}$ is unbiased.

(b)

Note that $Var(\bar{e}_1) = Var(\bar{e}_2) = \frac{\sigma^2}{n} = \frac{2\sigma^2}{N}$.

$$Var(\hat{\beta}_{2,\text{mean}}|\mathbf{x}) = \frac{Var(\bar{e}_2 - \bar{e}_1|\mathbf{x})}{(\bar{x}_2 - \bar{x}_1)^2} = \frac{\frac{2\sigma^2}{N} + \frac{2\sigma^2}{N}}{(\bar{x}_2 - \bar{x}_1)^2} = \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}$$

(c)

$$\text{Var}(\hat{\beta}_{2,\text{mean}}) = E \left[\frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2} \right] = \frac{4\sigma^2}{N} E \left[\frac{1}{(\bar{x}_2 - \bar{x}_1)^2} \right]$$

(d)

Consistency if $\lim_{N \rightarrow \infty} \text{Var}(\hat{\beta}_{2,\text{mean}}) = 0$

Require denominator $N(\bar{x}_2 - \bar{x}_1)^2 \rightarrow \infty$ as $N \rightarrow \infty$. That's, $(\bar{x}_2 - \bar{x}_1)$ does not vanish faster than $1/\sqrt{N}$.

(e)

i. ii.

Table 1: Simulation Results

	N	$\text{Var}(b_2 x)$	$\text{Var}(\hat{\beta}_{2,\text{mean}} x)$	$E[(s_x^2)^{-1}]$	$E[4/(\bar{x}_2 - \bar{x}_1)^2]$
1	100	1.244	1.663	0.124	0.166
2	500	0.244	0.326	0.122	0.163
3	1,000	0.123	0.166	0.123	0.166
4	5,000	0.024	0.032	0.121	0.162

(f)

As N increases, both $\text{Var}(b_2|\mathbf{x})$ and $\text{Var}(\hat{\beta}_{2,\text{mean}}|\mathbf{x})$ decrease at a rate of approximately $1/N$. However, the variance of the mean-based estimator is consistently larger than that of the OLS estimator. This efficiency gap arises because the scaling factor $E[4/(\bar{x}_2 - \bar{x}_1)^2]$ converges to approximately 0.16, whereas $E[(s_x^2)^{-1}]$ converges to a smaller value of approximately 0.12. Since the variance of $\hat{\beta}_{2,\text{mean}}$ shrinks to zero in the limit and the estimator is unbiased, we conclude that $\hat{\beta}_{2,\text{mean}}$ is a consistent estimator.

(g)

(1)

$$x \sim U(0, 10), \sigma_x^2 = \frac{(10 - 0)^2}{12} = 8.33$$

By the Continuous Mapping Theorem, if $s_x^2 \xrightarrow{p} 8.333$, then its inverse converges to the inverse of the limit:

$$(s_x^2)^{-1} \xrightarrow{p} \frac{1}{8.333} = \frac{12}{100} = 0.12$$

(2)

$$\bar{x}_1 \xrightarrow{p} \frac{0 + 5}{2} = 2.5$$

$$\bar{x}_2 \xrightarrow{p} \frac{5 + 10}{2} = 7.5$$

$$\bar{x}_2 - \bar{x}_1 \xrightarrow{p} 7.5 - 2.5 = 5$$

$$\frac{4}{(\bar{x}_2 - \bar{x}_1)^2} \xrightarrow{p} \frac{4}{5^2} = \frac{4}{25} = 0.16$$

(h)

No, the estimator $\hat{\beta}_{2,mean}$ would not be consistent if the observations on x were randomly assigned rather than ordered.

As $N \rightarrow \infty$, the difference $(\bar{x}_2 - \bar{x}_1) \xrightarrow{p} 0$. This violates the necessary condition for consistency discussed in part (d), which requires the total variation in the regressor to grow indefinitely.